AN EXACT COMPLEX OF LINE BUNDLES ON PROJECTIVE SPACE BUILT FROM THE KOSZUL COMPLEX

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Allowed standard prerequisites (editorial).

- The Koszul complex on a regular sequence is exact except in degree zero where its cohomology is the quotient by the sequence. invoked at proof lines 4344-4350. Provides the exactness of the Koszul complex K_bullet on the polynomial ring.
- Sheafification on Proj is exact and kills modules supported in positive degree such as the quotient by the coordinates. invoked at proof lines 4353-4361. Turns the exact complex of graded modules into an exact complex of vector bundles ending at O_X.

Source reference. Original definition 01NF: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4339-4341 for the identification of P^n_A with Proj of the polynomial ring; the Koszul construction on it is inside the counted span.

Source reference. Original lemma 062F: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4346-4350 for the exactness of K_bullet except in degree 0; the grading and sheafification of the resulting complex are inside the counted span.

Source reference. Original lemma 01M7: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4357-4361 to sheafify the graded complex and drop the quotient term; the twist concluding the proof is inside the counted span.

Lemma 0.2. *Let A be a ring. Let $X = \mathbf{P}^n_A$. For every $a \in \mathbf{Z}$ there exists an exact complex*

$$0 \to \mathcal{O}_X(a) \to \ldots \to \mathcal{O}_X(a+i)^{\oplus \binom{n+1}{i}} \to \ldots \to \mathcal{O}_X(a+n+1) \to 0$$

of vector bundles on X.

Proof. Recall that $\mathbf{P}^n_A$ is $\mathrm{Proj}(A[X_0, \ldots, X_n])$, see Constructions, Definition 01NF. Consider the Koszul complex

$$K_\bullet = K_\bullet(A[X_0, \ldots, X_n], X_0, \ldots, X_n)$$

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over $S = A[X_0, \ldots, X_n]$ on $X_0, \ldots, X_n$. Since $X_0, \ldots, X_n$ is clearly a regular sequence in the polynomial ring S, we see that (More on Algebra, Lemma 062F) that the Koszul complex $K_\bullet$ is exact, except in degree 0 where the cohomology is $S/(X_0, \ldots, X_n)$. Note that $K_\bullet$ becomes a complex of graded modules if we put the generators of K_i in degree $+i$. In other words an exact complex

$$0 \to S(-n-1) \to \ldots \to S(-n-1+i)^{\oplus \binom{n}{i}} \to \ldots \to S \to S/(X_0, \ldots, X_n) \to 0$$

Applying the exact functor $\tilde{}$ functor of Constructions, Lemma 01M7 and using that the last term is in the kernel of this functor, we obtain the exact complex

$$0 \to \mathcal{O}_X(-n-1) \to \ldots \to \mathcal{O}_X(-n-1+i)^{\oplus \binom{n+1}{i}} \to \ldots \to \mathcal{O}_X \to 0$$

Twisting by the invertible sheaves $\mathcal{O}_X(n+a+1)$ we get the exact complexes of the lemma. $\square$